PLAIN-ENGLISH GUIDE · 2026

Explained Simply: Compounding

The $24 magic money tree: how small money grows huge.

A story about an island, an apple seed and a snowball, and what it teaches us about time and money. With charts, real numbers and the things that can break the magic.

- The $24 island
- The doubling trick
- Arjun and Saachi
- What breaks it
- Passing it down
- ISAs and pensions

MoneyBySharma

General information, not financial advice. Examples are illustrative, not forecasts.

THE STORY

The island that cost $24

How a tiny price became a famous money lesson

In 1626, settlers from the Netherlands got the island of **Manhattan**. Today it is the busy heart of New York City.

What they paid

- The story says **60 guilders** (Dutch money of the time).
- Later, people worked out this was about **$24**.
- Today, $24 would not even buy one new video game.

What it is now

- One of the most expensive places on Earth.
- Nobody knows the true price for sure.
- Historians still argue about it.

THE POINT OF THE STORY

A man named Mohnish Pabrai likes to tell this story. His point is not about land. It is about a superpower called compounding.

Jargon check: an **investor** is someone whose job is growing money by putting it into things like businesses.

IN THIS GUIDE

What compounding is, what $24 could have become, the doubling trick, what it could mean for you, and **what breaks the magic**.

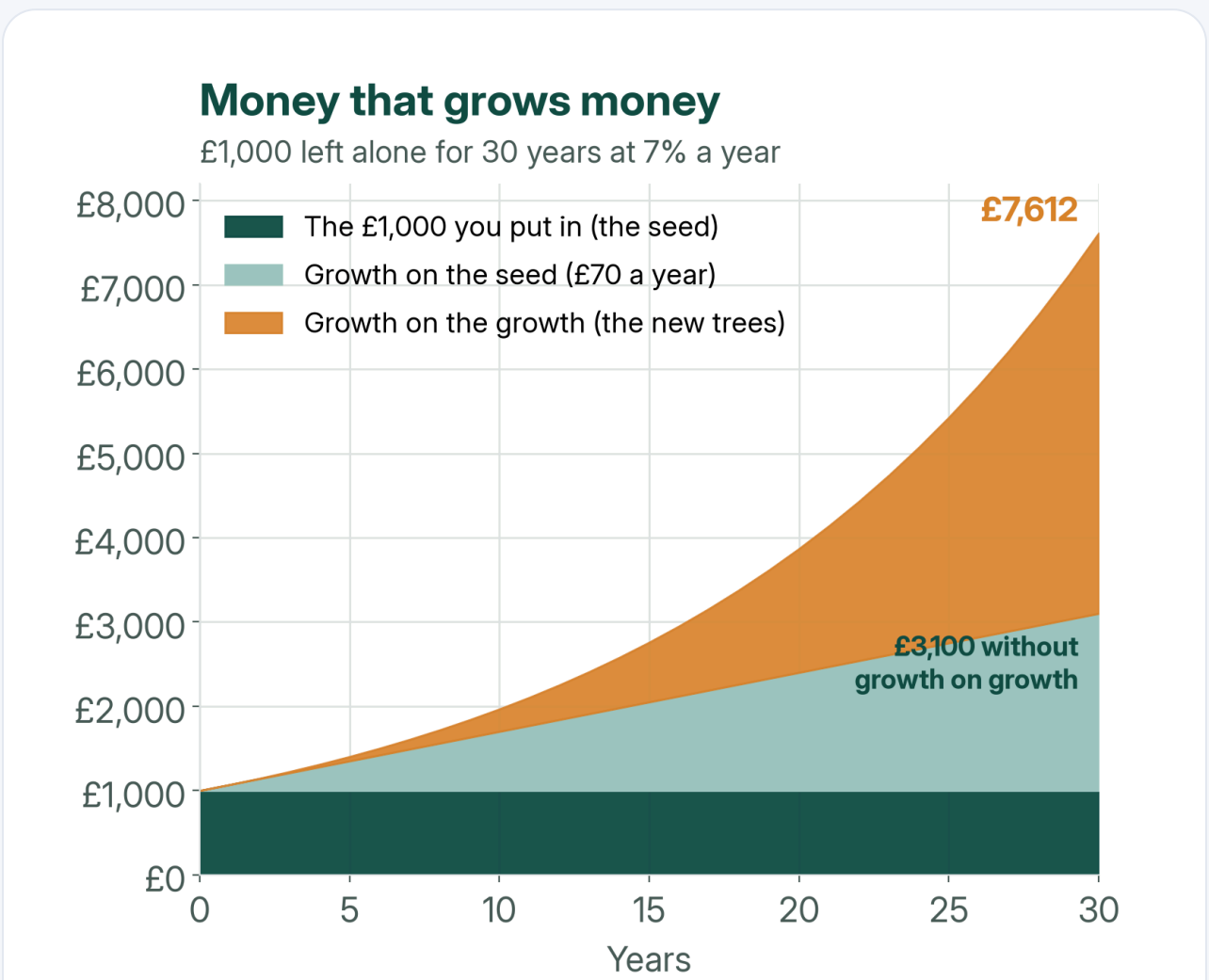
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THE IDEA

What is compounding?

Growth on top of growth

Plant an apple seed and it grows into a tree. The tree gives apples. Plant their seeds and you get more trees. **Money can work the same way:** it earns money, and next year that new money earns money too.



£1,000 at 7% a year for 30 years: about **£7,600** with compounding, against about £3,100 with simple growth.

REMEMBER

Compounding is money having babies, and **the babies having babies**. Like a snowball: the bigger it gets, the faster it grows.

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THE NUMBERS

The $24 at 7% a year

If it had been invested in 1626 and never touched

Seven percent means that for every $100, you earn **$7 each year**. That is not a crazy-high number. But it is **not guaranteed** either.

After	Year	About how much	How big is that?
Start	1626	**$24**	Less than one new video game
100 years	1726	**$20,800**	A used car
200 years	1826	**$18 million**	A giant lottery prize
300 years	1926	**$15.7 billion**	Billions!
400 years	2026	**$13.6 trillion**	More than almost every country makes in a year

The surprising part: for the first 100 years, not much seems to happen. In the **last 100 years alone**, the money grows by more than $13 trillion.

HOW IT GROWS

Slow, slow, slow... **then WHOA!**

Worked example: 7% a year, added once a year, never taken out. No tax, fees or inflation.

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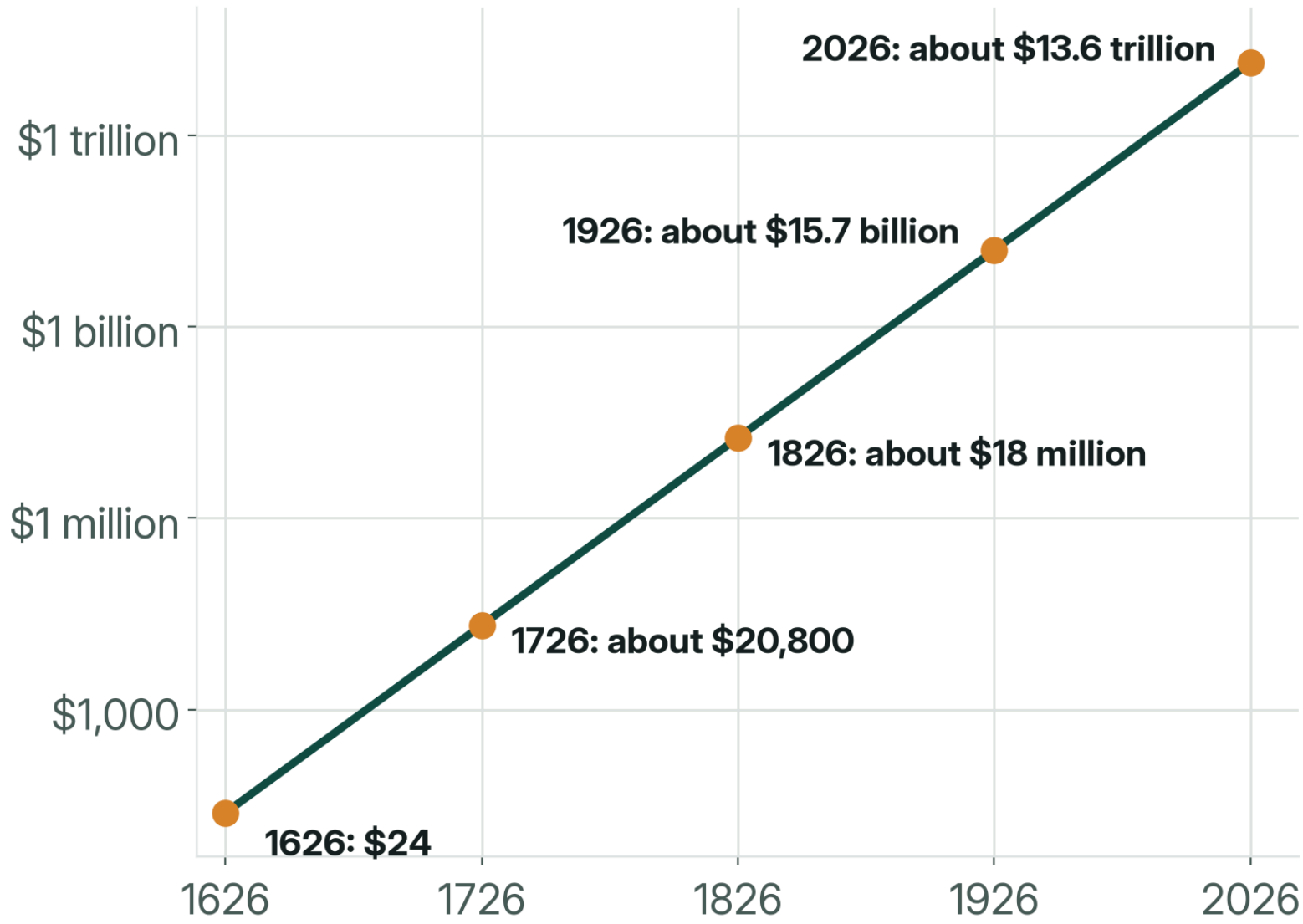
THE CHART

Slow, slow... then WHOA!

400 years of growth on one chart

What if the $24 grew at 7%?

Each step up the scale is 1,000 times bigger



Pretend example: $24 invested in 1626 at 7% a year, never touched. Log scale, so a straight line means steady growth. It shows how the maths works; it does not predict the future.

Each line up the side is **1,000 times** bigger than the one below. That is the only way to fit $24 and $13.6 trillion on one chart.

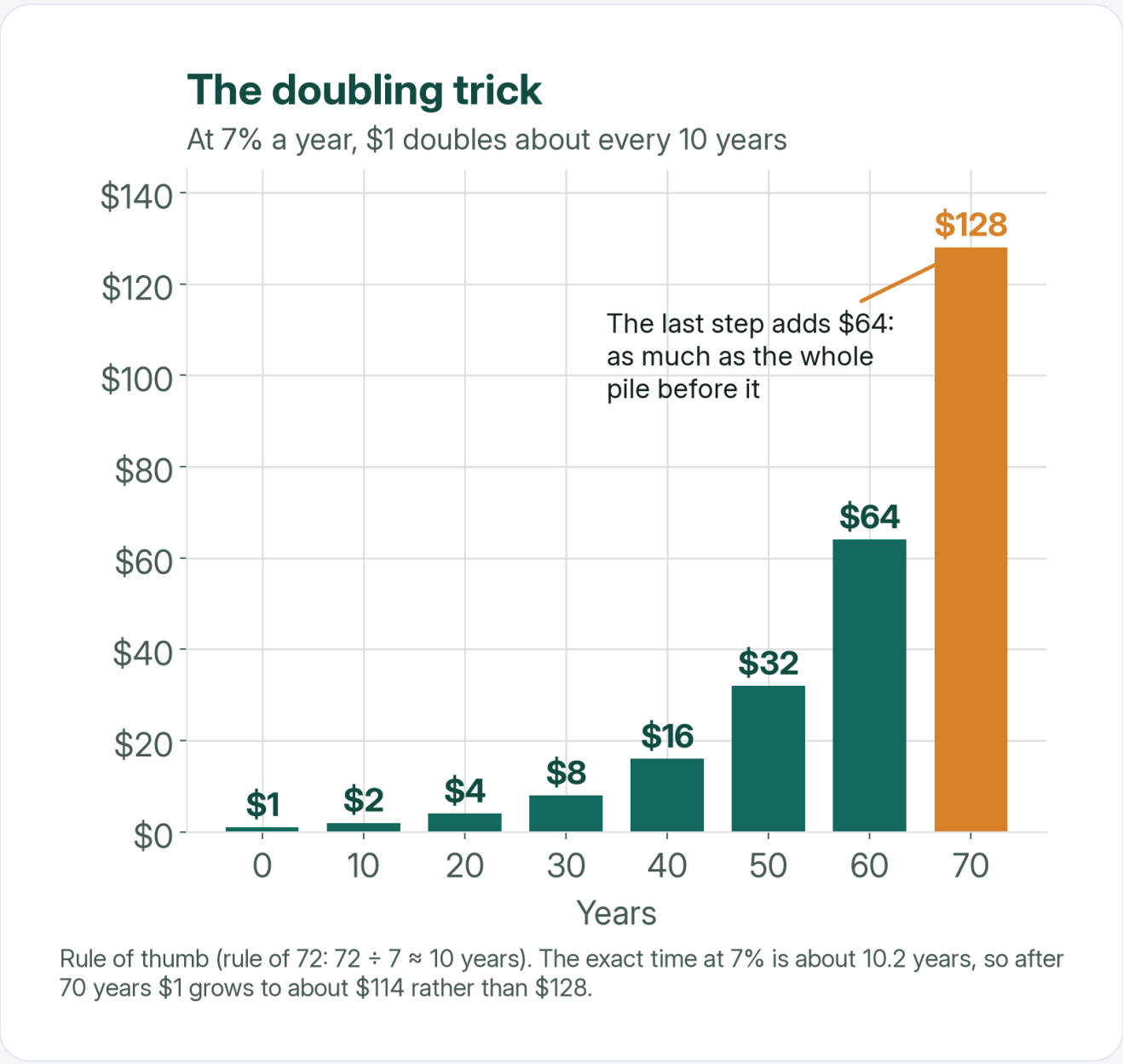
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THE SHORTCUT

The doubling trick

The rule of 72

At 7% a year, money **doubles about every 10 years**. Quick check: divide 72 by the yearly growth rate. **72 ÷ 7 ≈ 10 years.** At 4%, it would be 72 ÷ 4 = about 18 years.



Using the “about 10 years” rule. The exact figure after 70 years at 7% is about $114.

THE SECRET INGREDIENT IS TIME

Each step adds as much as **the whole pile before it**. That is why the end is the most exciting part.

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A REGULAR PERSON

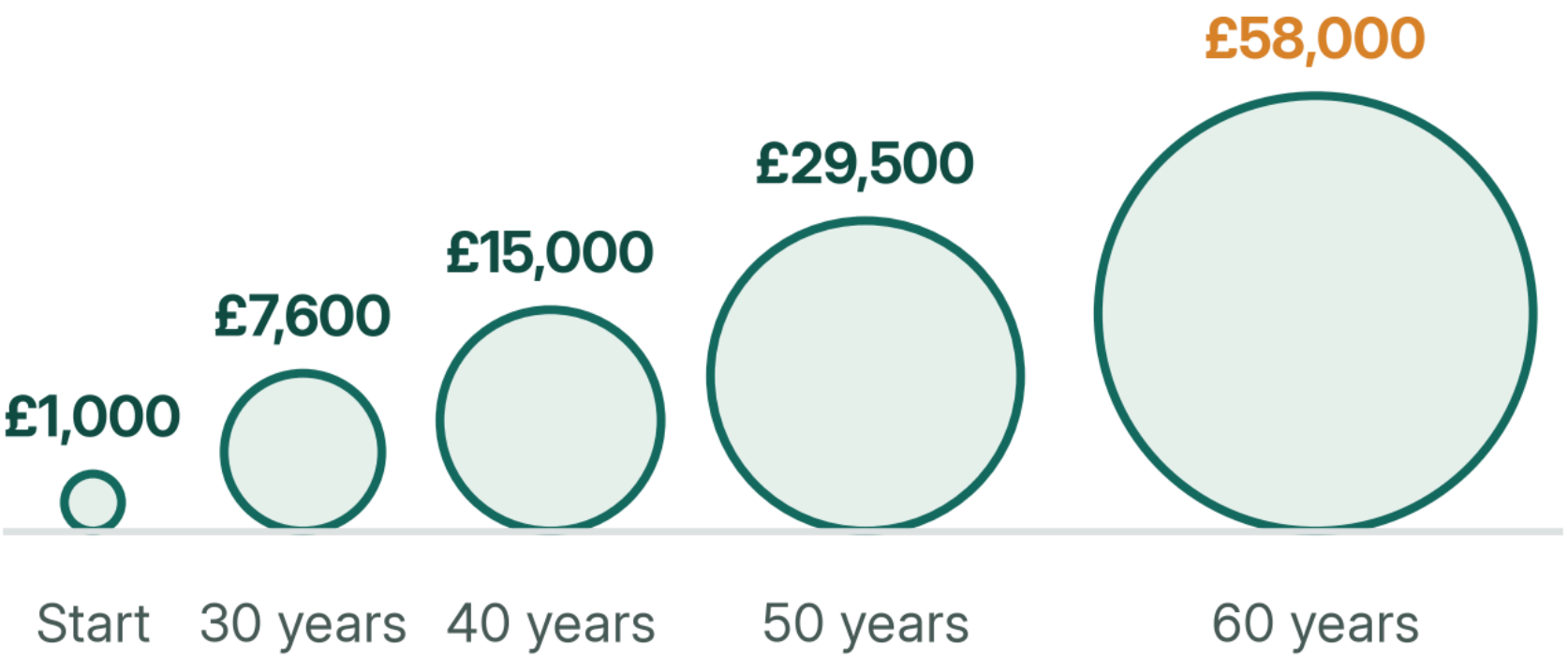
A £1,000 gift, left alone

Nobody lives 400 years, but you might invest for 50 or 60

Imagine someone gives you **£1,000** and it grows by 7% a year, never touched.

After	About how much
30 years	£7,600
40 years	£15,000
50 years	£29,500
60 years	£58,000

The snowball: a £1,000 gift, never touched



Illustrative only. 7% a year, added once a year, no fees, tax or inflation.
Each ball's area matches its value.

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START EARLY

Arjun and Saachi

Same savings, different start

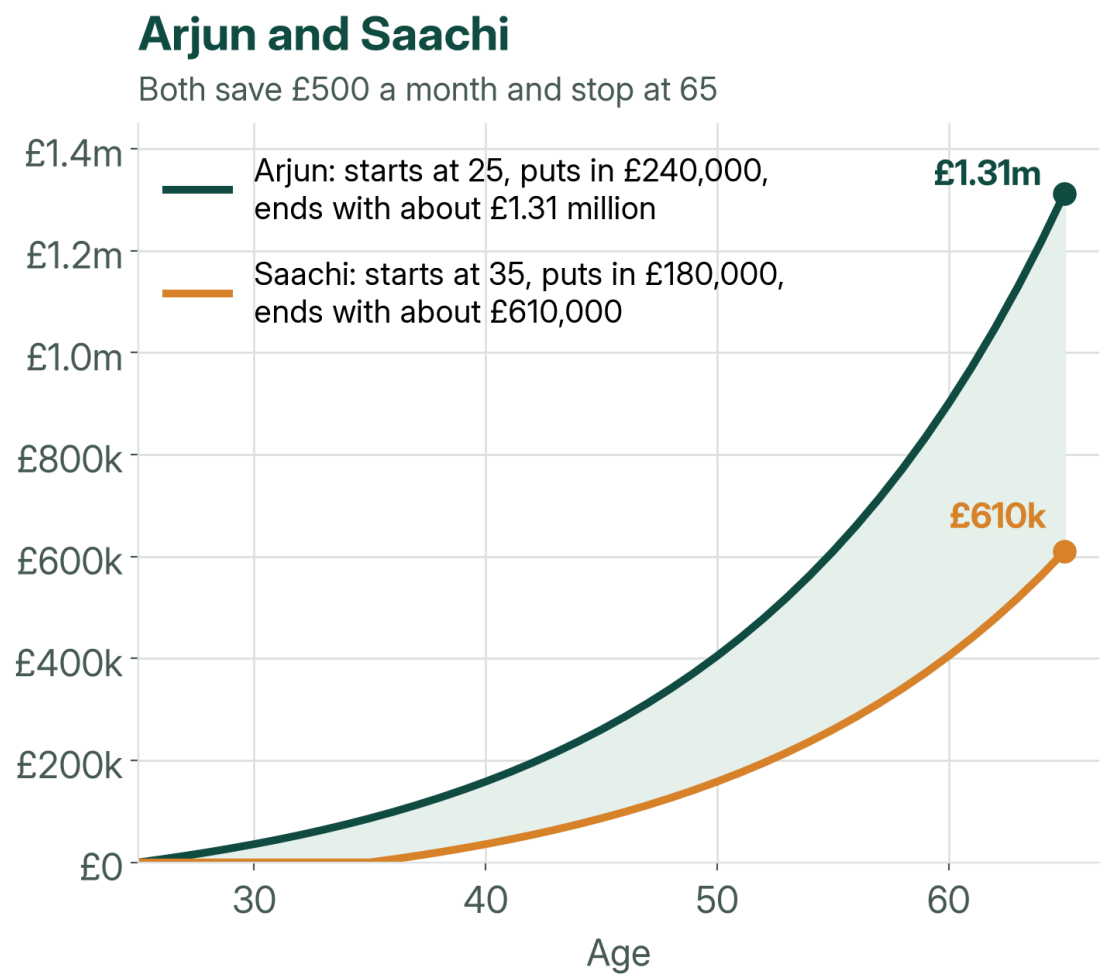
Both save **£500 a month** and stop at 65. Arjun starts at 25. Saachi starts at 35.

ARJUN · STARTS AT 25

Puts in £240,000.
Has about **£1.31 million**.

SAACHI · STARTS AT 35

Puts in £180,000.
Has about **£610,000**.



Illustrative only. 7% a year, compounded monthly, £500 added at the end of each month. No fees, tax or inflation. Real returns go up and down.

STARTING EARLY IS A SUPERPOWER

Saachi waited just 10 years and ended up with **about £700,000 less**, despite paying in only £60,000 less.

Worked example: 7% a year, added monthly, £500 paid in at the end of each month. No fees, tax or inflation.

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WATCH OUT

What breaks the magic

Three things that stop the tree growing

Compounding needs **time** and needs to be **left alone**. Three things get in the way.

1. Starting late

The last years are the biggest ones, so **every year you wait costs you a lot**. Arjun and Saachi show how much.

2. Big losses

Losing money hurts more than you think. **Lose half and you need to double** what is left just to get back to where you started.

3. Taking it out early

If you dig up the tree to check the roots, it stops growing. **High fees do the same thing**, because they nibble away at the growth.

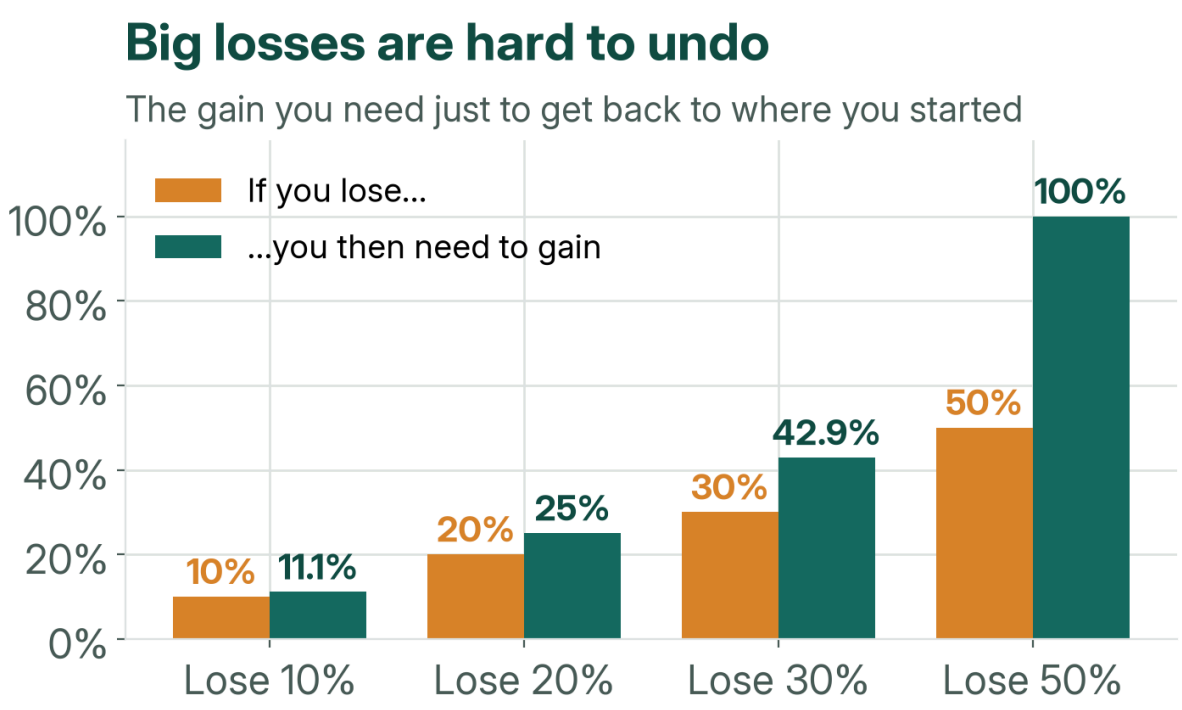
Example: £10,000 left for 30 years grows to about **£76,100** at 7% a year. A 1% yearly fee (so 6%) leaves about **£57,400**, roughly £18,700 less.

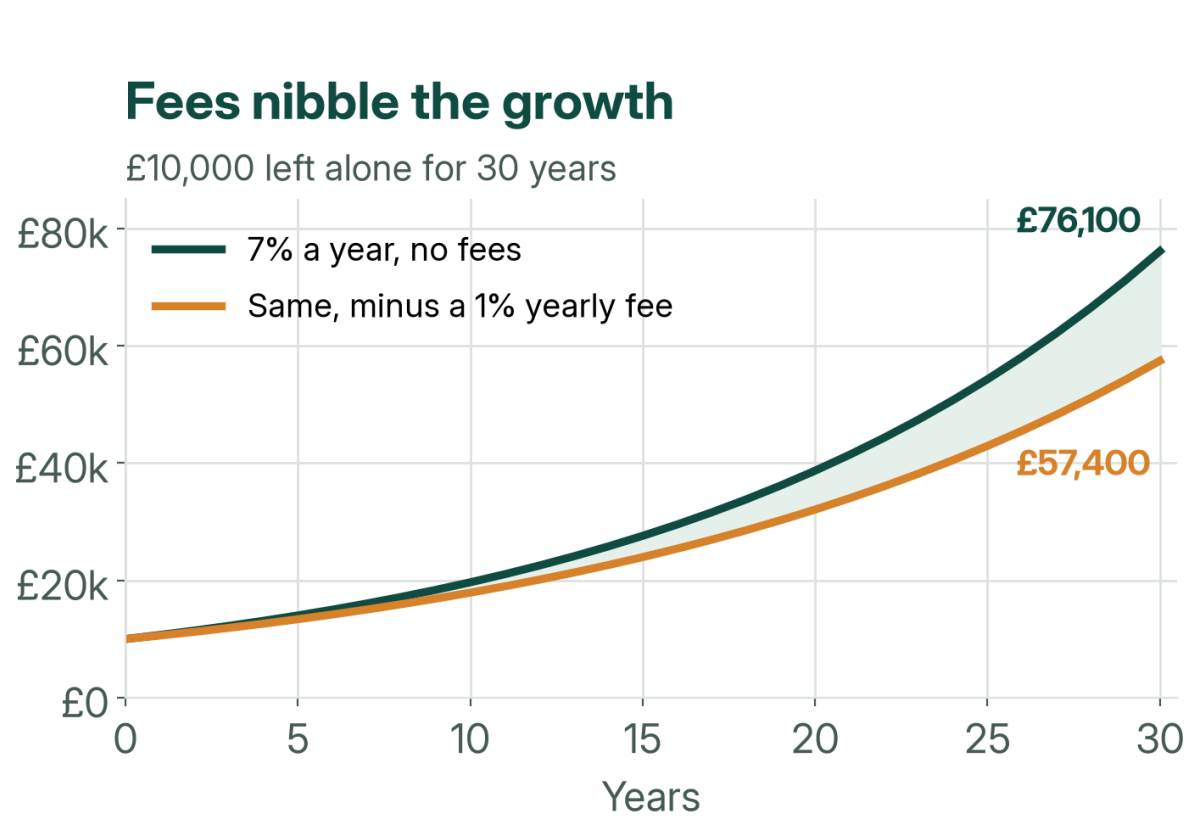
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IN PICTURES

Losses and fees

Why they matter so much





Illustrative only. Growth added once a year. The fee is shown as 1 percentage point off the return (7% becomes 6%). No tax or inflation. Real returns go up and down.

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GENERATIONS

Passing it down

Generational wealth

Generational wealth means money that helps your children, and their children too. You plant the tree, but your grandchildren might sit in its shade.

£1 MILLION AFTER 30 YEARS

About **£7.6 million**

£1 MILLION AFTER 60 YEARS

About **£58 million**

What long-lasting wealthy families usually do

- **Teach the children.** A child who understands compounding starts with a huge head start.
- **Spend less than the money grows.** If it grows 7% and they take out 4%, the tree keeps getting bigger.
- **Keep costs low.** Smaller fees mean more growth stays in the tree.
- **Own things that grow**, like shares (small pieces of businesses), not just cash under the mattress.

Worked example: 7% a year, added once a year. No tax, fees or inflation.

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REAL MARKETS

7% vs real stock markets

What three famous share indexes did over about 40 years

All our examples use **7% a year**. Here is what three famous share indexes returned on average, with **dividends** (the share of profits companies pay out) put back in.

Index	What it is	Period	A year
FTSE 100	The 100 biggest companies on the London Stock Exchange	1984 launch to end of 2023, in £	**8.10%**
S&P 500	About 500 large US companies	1985 to 2024, in US$	**11.68%**
Nasdaq-100	100 large non-financial US Nasdaq companies, many in tech	Jan 1985 launch to end of 2024, in US$	**14.25%**

Same method as Arjun and Saachi: £500 a month, paid in at the end of each month, growing monthly, until 65.

Rate a year	Arjun ends with	Saachi ends with	Gap
7.00% Our example	**£1,312,000**	**£610,000**	£702,000
8.10% FTSE 100	**£1,797,000**	**£760,000**	£1,036,000
11.68% S&P 500	**£5,318,000**	**£1,628,000**	£3,690,000
14.25% Nasdaq-100	**£12,126,000**	**£2,909,000**	£9,217,000

Worked example, rounded to the nearest £1,000. Arjun pays in £240,000 from 25; Saachi pays in £180,000 from 35. No fees, tax or inflation, and the same rate every year, which never happens in real life.

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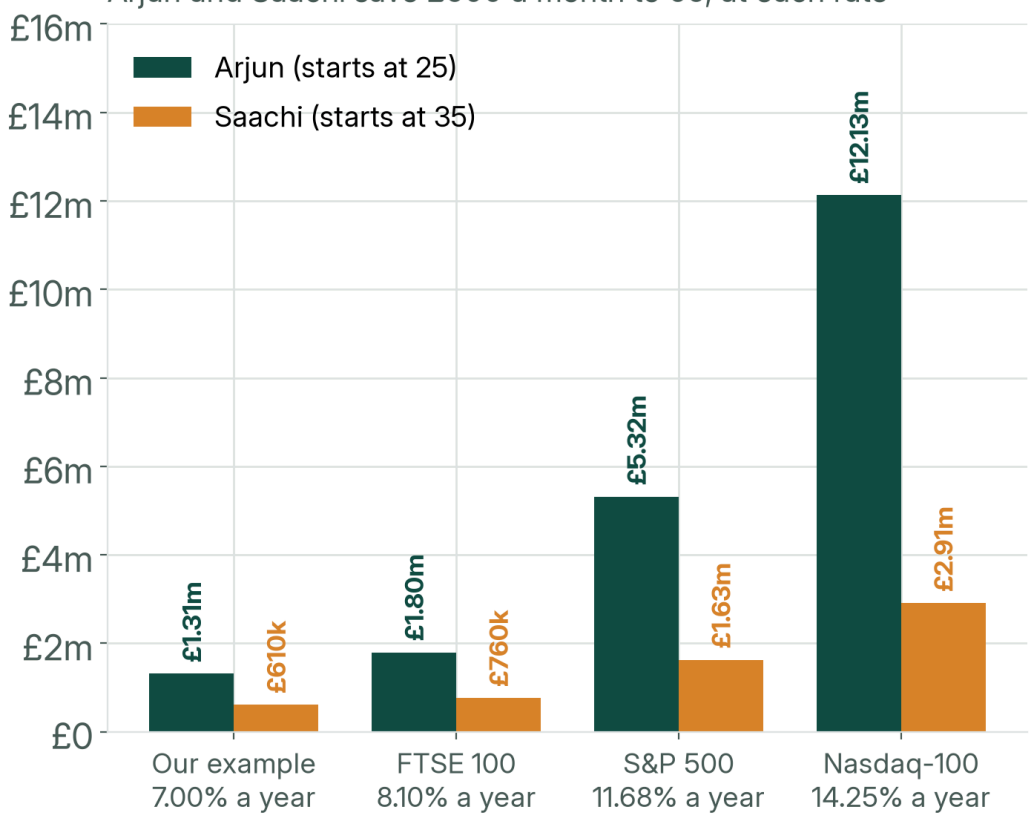
REAL MARKETS

Arjun & Saachi, real rates

Same £500 a month, four different rates

7% vs real market averages

Arjun and Saachi save £500 a month to 65, at each rate



Illustrative only. Past ~40-year average yearly returns with dividends reinvested, before fees, tax and inflation (S&P 500 and Nasdaq-100 in US dollars, FTSE 100 in pounds). Compounded monthly, £500 added at the end of each month. Past returns are not a promise of future returns.

⚠ Before you get excited

- **Exchange rates matter.** S&P 500 and Nasdaq-100 returns are in US dollars. A UK investor's returns would also move with the pound-to-dollar rate.
- **Past averages, not a promise.** The Nasdaq-100 fell 83% when the tech bubble burst around 2000.
- **Fees, tax and inflation** reduce what you really keep.

Sources: FTSE 100: The Guardian, "Footsie turns 40", 2 Jan 2024 (£1,000 at the 1984 launch grew to £22,550 with dividends reinvested; 8.10% a year). S&P 500: A. Damodaran, NYU Stern, "Historical Returns on Stocks, Bonds and Bills", data updated Jan 2026 (S&P 500 incl. dividends, 1985–2024; 11.68% a year). Nasdaq-100: Nasdaq, "Nasdaq-100 Index Celebrates 40 Years of Innovation", 30 Jan 2025 (14.25% a year, 31 Jan 1985 to 31 Dec 2024). All nominal, before fees, tax and inflation.

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BE HONEST

7% is not a promise

What the story does and doesn't tell you

Returns go up and down

Some years the money goes down, not up.

Prices rise

This is called **inflation**. Money buys a little less each year.

Nobody invests 400 years

The Manhattan story shows how the maths works. It does not predict the future.

Information, not advice

Before making decisions, speak to a **qualified financial adviser** (one authorised by the FCA).

A UK note: ISAs and pensions

UK

- Money invested inside an **ISA** or a **pension** grows without being taxed as it builds up.
- That means more of the growth stays in the pot to **keep compounding**.
- Pension money is usually taxed when you take it out, apart from a tax-free part.
- The rules and limits can change.

Your money is at risk. When you invest, you could get back less than you put in.

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The big idea

A small amount, a steady return and **lots of time** can beat a big amount that gets interrupted. You can control when you start, how much you add and how long you leave it alone. **Plant the seed early, protect it, and let time do the rest.**

72 ÷ 7 ≈ 10
Years to double at 7% a year

10 years
Arjun's head start was worth about £700,000

£1,000 → £58,000
A gift left alone for 60 years at 7% a year

Lose 50%
...and you need +100% just to get back

Think about it
If you planted £24 today, about how old would you be when it doubled? (Hint: about 10 years from now.)

Share it
With your family, so the next generation learns it early.

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